

# **Module 10**

## **Allocating Costs to Jobs and Processes**

## Learning Objectives

**By the end of this module you should understand:**

1. The sources for costs of direct material, direct labor and overhead in the cost gathering process
2. The use of cost centers to gather overhead costs
3. The application of activity bases to spread overhead to products
4. The distinction between plant wide and departmental predetermined overhead rates
5. The costing issues involved in joint products and by-products
6. Process costing, its use of equivalent units and its two principal methods
7. The need to question the appropriateness of traditional cost drivers and the basic mechanisms of activity-based costing

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10.4 Plantwide versus Departmental Rates

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10.5 Joint Products and By-Products

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10.6 Process Costing

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## In module 9 we've covered different mutually exclusive cost categories

### Cost Categories:

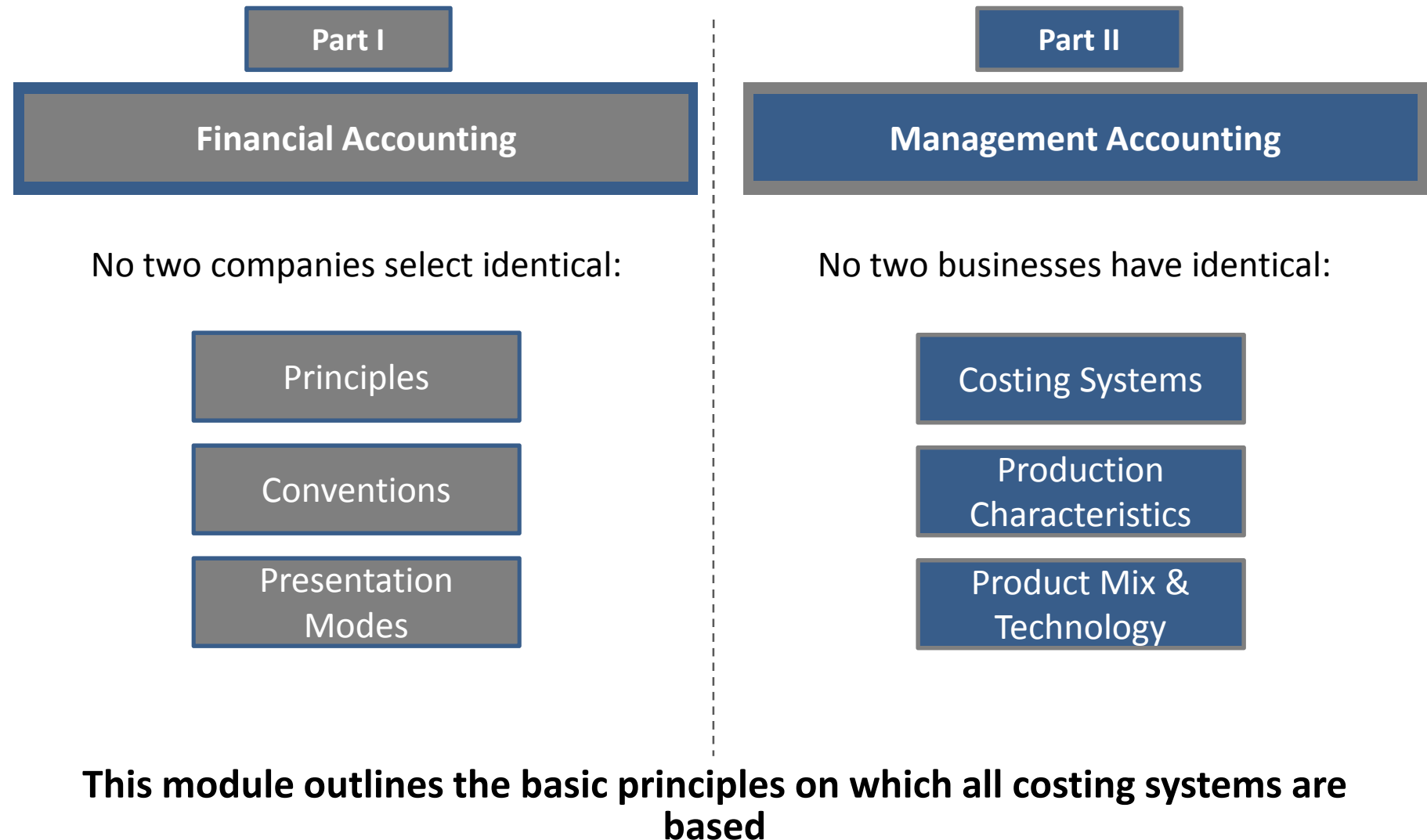


**These categories are broad as they deal with total blocks of costs**



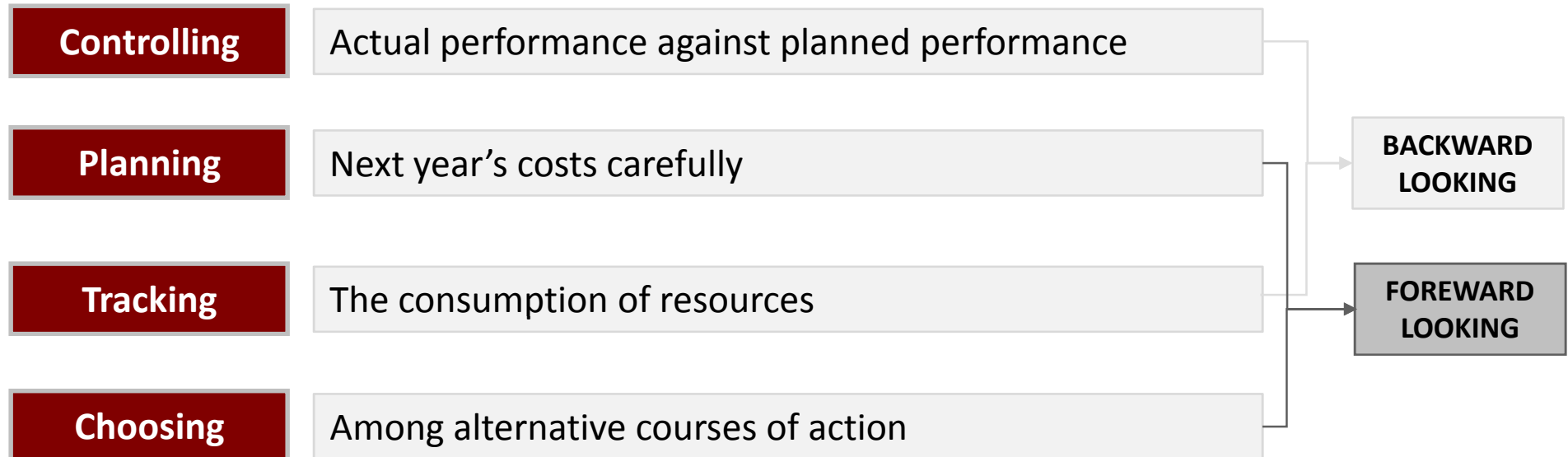
**An early understanding of their characteristics allows the reader to gain a broad overview of costing and its potential impact on managerial behavior**

**No two businesses have identical costing systems because no two companies have identical accounting systems**



As discussed in the previous module, costing systems have 4 roles: two of them are forward looking and two are backward looking

### Why is Cost Information so Important?



This module will concentrate on the **tracking** and **controlling** functions of cost systems



**When managers are satisfied with those two functions, they'll be able to assess their informational needs for planning and choosing**

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**To focus our discussion on cost gathering (also known as cost accumulation), we'll use the same example of a timber-shelved DVD unit used in Module 8**

**One timber-shelved DVD unit**

|                                       | €           |
|---------------------------------------|-------------|
| Direct material:                      |             |
| Chipboard                             | 10.00       |
| Veneer                                | 3.00        |
| Metal trim                            | 2.00        |
|                                       | <hr/>       |
|                                       | 15.00       |
| Direct labour 1.5 hours @ €5 per hour | 7.50        |
|                                       | <hr/>       |
| Prime cost                            | 22.50       |
| Manufacturing overhead                | 2.50        |
|                                       | <hr/>       |
| Manufacturing cost                    | 25.00       |
| Share of non-manufacturing overhead   | 5.00        |
|                                       | <hr/>       |
| Total cost                            | <hr/> 30.00 |

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## Three items we have to consider when tracing costs



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**Materials are the easiest cost to trace to a cost item because of their visible, physical nature; where its percentage from total costs varies from item to item**

|           | Item of furniture | Personal computer | Software package |
|-----------|-------------------|-------------------|------------------|
|           | %                 | %                 | %                |
| Materials | 50                | 25                | 2                |
| Labour    | 25                | 25                | 8                |
| Overhead  | 25                | 50                | 90               |
|           | <u>100</u>        | <u>100</u>        | <u>100</u>       |

*Heavy overhead component in the software package reflects the massive investment in **R&D** required by a software house to design and test a commercial package*

**Materials are drawn from inventory where a requisition slip is used to record the quantity of materials drawn, their quality materials, the date drawn and the initials of the storekeeper and the requisitioner**

## Determining the price of material is not an easy task; let's see why in this example

### Example

The metal trim required in the construction of the DVD units was purchased in five meter lengths from a local steel stockholder. Varying quantities were purchased during the month of May at various prices.

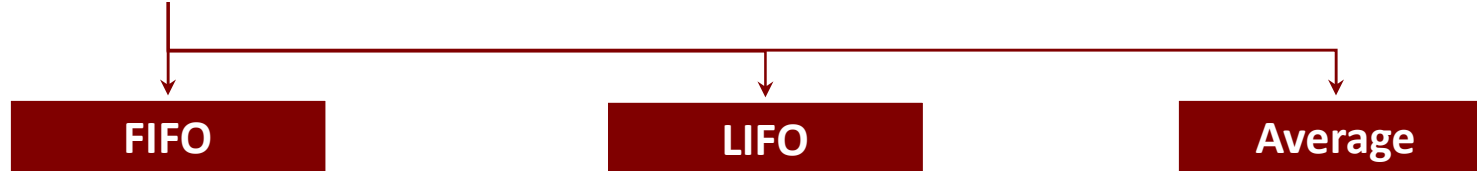
#### *May*

|    |                              |
|----|------------------------------|
| 3  | 5 metres @ €0.90 per metre   |
| 16 | 90 metres @ €0.96 per metre  |
| 21 | 10 metres @ €0.89 per metre  |
| 30 | 25 metres @ €1.06 per metre. |

**Assume that the metal storekeeper issued 7 meters of metal to production each working day of the month. What price should be charged?**

## Solution

A number of possibilities suggest themselves:



However, most likely they'd set:

### Standard Price

For, perhaps, a three-month period and which would attempt to predict the day-by-day price fluctuations in the market

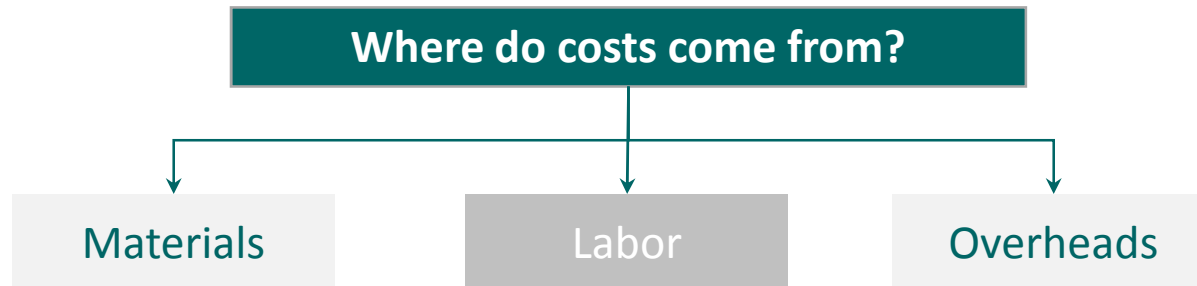
Any differences which arise between actual purchase price and production issuing price are accounted for in the P&L account as a price variance at the end of the given period

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### Just In Time

Manufacturing companies encourage individual departments to order their parts direct from the supplier ***just in time*** for using in the production cycle; reducing inventories of raw materials to almost zero

## Three items we have to consider when tracing costs



**Labor is also a relatively easy cost to identify and measure, particularly if the persons involved are directly involved in the construction of the cost item**

$$\begin{array}{c} \text{Labor cost} \\ \\ = \\ \text{Time spent on a job} \\ \\ \times \\ \text{Rate of pay earned by the operatives involved} \end{array}$$

**Each operative may maintain a daily time sheet on which to record the jobs worked on and the time per job**

**Rate per hour is not easy to determine; operatives, because of differing skill levels or years of service, earn different rates of pay**

### **Example**

One timber-shelved unit may be worked on by two operatives for 45 minutes each; one of these persons earned €4.60 per hour, the other €5.00 per hour. Simultaneously, two other operatives, earning €5.10 and €5.15 per hour respectively, spent 45 minutes constructing another unit.

**Is it fair to charge different rates of pay to identical products?**

### **Solution**

No, it is not fair

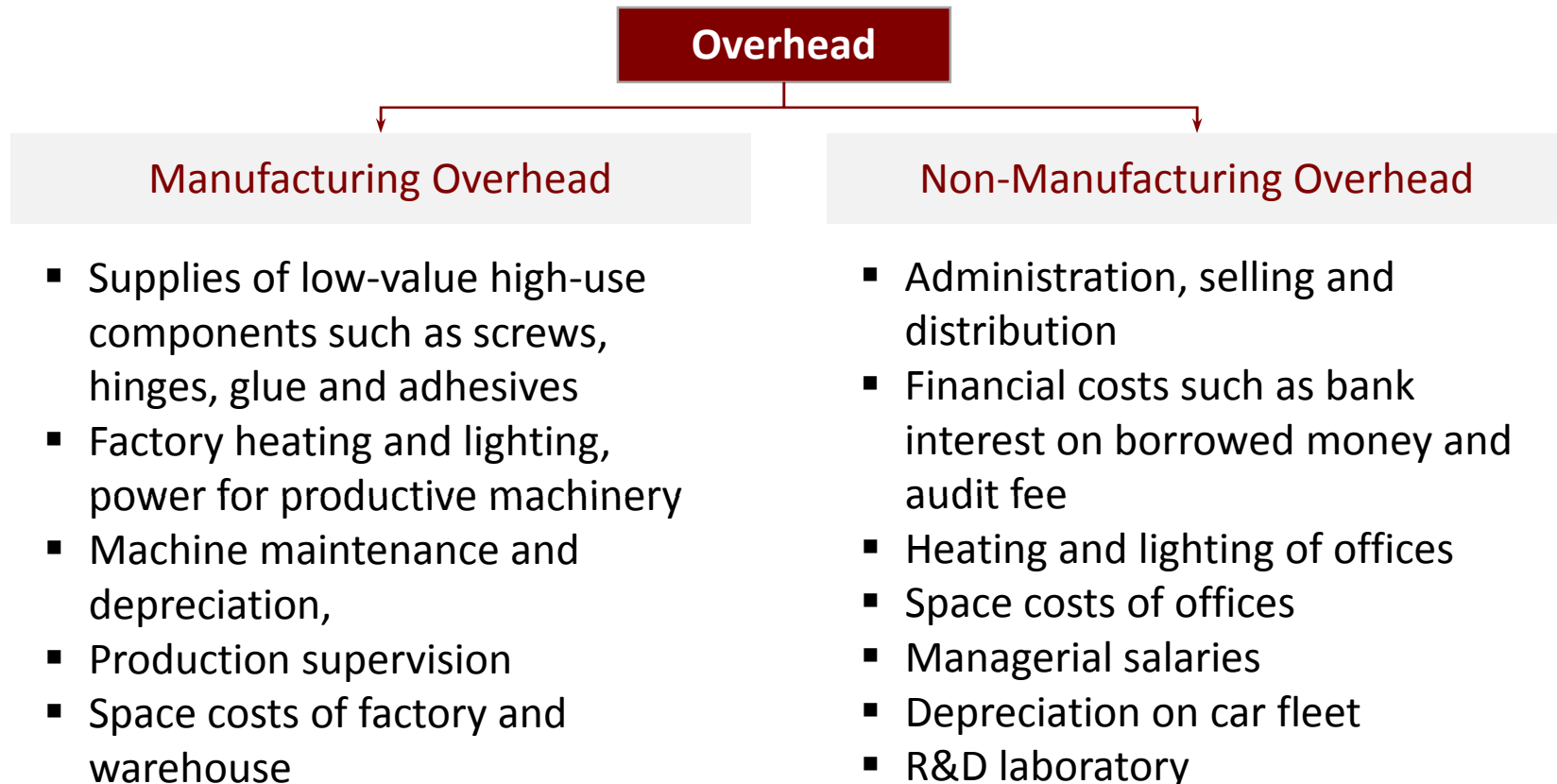
*we use*  
→

**Standard Wage Rate**

## Three items we have to consider when tracing costs



**Before spreading the overheads to units of production these costs must be at first gathered & controlled; this is hard since they are not directly linked to output**



**These costs have a tendency to escalate beyond a level which can be justified by either the business's production or profit**

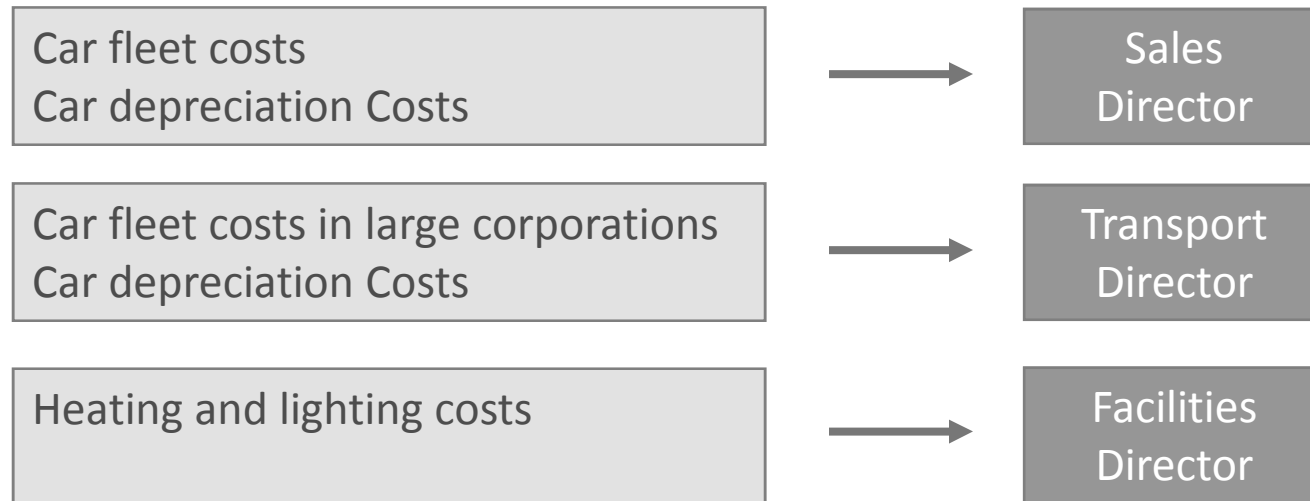


*that's why some see them as*

**Non-Controllable Costs**

**Overhead costs should be gathered initially into ‘*cost centers*’ – pockets of activity for which individual managers may be held responsible**

### **Costs Centers’ Managers**



**Each principal cost centre manager may decide to split his or her centre into sub-units of responsibility and control where a target level of expenditure will be set for each sub-unit beyond which managers must not go**

Then comes the task of allocating these costs to units of production which is far harder than the straightforward task of cost gathering

### Example

One timber-shelved DVD unit

|                                       | €     |
|---------------------------------------|-------|
| Direct material:                      |       |
| Chipboard                             | 10.00 |
| Veneer                                | 3.00  |
| Metal trim                            | 2.00  |
|                                       | <hr/> |
|                                       | 15.00 |
| Direct labour 1.5 hours @ €5 per hour | 7.50  |
|                                       | <hr/> |
| Prime cost                            | 22.50 |
| Manufacturing overhead                | 2.50  |
|                                       | <hr/> |
| Manufacturing cost                    | 25.00 |
| Share of non-manufacturing overhead   | 5.00  |
|                                       | <hr/> |
| Total cost                            | 30.00 |

*How were the amounts of €2.50 and €5.00 per DVD cabinet calculated for, manufacturing overhead and non-manufacturing overhead?*

*How much depreciation of production machinery was included in the €2.50?*

*How much managerial Salaries were included in the €5.00?*

**These overheads are not directly attributable to cost units in the same manner as direct materials and direct labor**

**Accountants use a predetermined overhead rate to spread overheads across the units of production**

Predetermined Overhead Allocation Rate

=

$$\frac{\text{Budgeted overhead for accounting period}}{\text{Budgeted Production Units}}$$

## Example

A business manufactures and sells one product which contains €5 direct material and €10 direct labor. Over the forthcoming year management plans to manufacture 10,000 units of the product and the estimate of total overhead amounts to €100,000.

The predetermined overhead allocation rate would be:

$$\frac{€100\,000}{10\,000} = €10 \text{ per unit}$$

This rate would then be slotted into the cost profile of the product on which pricing and marketing decisions would be taken:

| Cost profile for product |            |
|--------------------------|------------|
| Direct material          | € 5        |
| Direct labour            | 10         |
| Share of overhead        | 10         |
| Total cost               | <u>€25</u> |

## However the simple fraction does not work for a business which produces more than one product with different production characteristics

If the above business made 9,000 units of the existing product and 2,000 units of a related, but different, product.

It would be ridiculous to use the figure of 11,000 for the denominator of the fraction. The business must use some other criterion to spread cost.

↓  
*That's why:*

**Cost accountants search for *causal factors* or *activity bases*, that cause the majority of the overheads**

Labor-intensive business

Automation

**Labor-hours is a useful causal factor**

**Machine-hours is a useful causal factor**

**The resulting predetermined rate is applied to the individual products using these products' consumption of the causal factor**

**In planning, businesses usually examine closely last year's performance and consider management's expectations to reduce or increase specific overhead outlays**

### **Example**

In the previous example assume that direct labor hours is to be used in the predetermined overhead allocation rate calculation, that the business plans to spend 25,000 direct labor hours on production next year and that this product (one of several) takes 2 hours to construct by direct labor staff.

$$\begin{aligned}\text{Predetermined overhead allocation rate} &= \frac{\text{€100 000}}{25\,000 \text{ direct labour hours}} \\ &= \text{€4 per direct labour hour}\end{aligned}$$

| Cost profile for product         |            |
|----------------------------------|------------|
| Direct material                  | € 5        |
| Direct labour                    | 10         |
| Share of overhead (2 hours × €4) | 8          |
|                                  | <u>€23</u> |

**The difference between the actual figure of overhead incurred and that estimated to the product is accounted directly to the P/L account**

**This difference (the under-applied amount in that example) would be charged directly to the P/L account as an increase to the figure for the cost of goods sold**

### **Example**

At the end of the year the business above incurred €105,000 on overhead costs. Production was equivalent to 26,000 direct labor hours.

|                                            |               |
|--------------------------------------------|---------------|
| Overhead applied to products = 26 000 × €4 | €104 000      |
| Actual overhead                            | €105 000      |
| Underapplication                           | <u>€1 000</u> |

**The cost accountants would investigate the cause of this overspend and overproduction so that their figures for next year would be even more accurate**

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**Management must design a cost-allocation system that reflects each product's actual usage of overheads rather than using the blunt instrument of a plantwide rate**

**Typical Activity Bases For Service Department Costs Include the Following:**

|                                        |                                        |                                  |                                 |
|----------------------------------------|----------------------------------------|----------------------------------|---------------------------------|
| <b>Personnel/<br/>Welfare Services</b> | Number of persons<br>employed          | <b>Power</b>                     | Machine-hours;<br>metered usage |
| <b>Computing</b>                       | hours run; number<br>of reports issued | <b>Executive<br/>Salaries</b>    | Sales                           |
| <b>Machinery</b>                       | Machine-hours                          | <b>Production<br/>Scheduling</b> | Number of different<br>products |
| <b>Buildings and<br/>Estate</b>        | Space occupied                         |                                  |                                 |

**The plantwide predetermined overhead rate described before, used either by a one-product firm or a more-than-one-product firm using only one activity base is unrealistic**

## Departmental overhead rates attempt to avoid the problem of the differing activity bases

First

Through allocating overheads to individual departments

Second

Through identifying the activity base most appropriate for each department

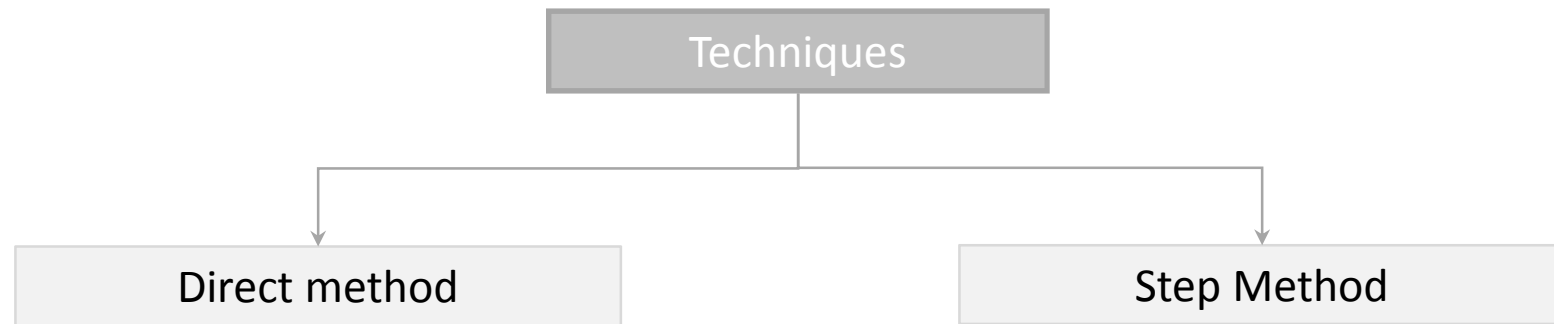
## We'll use the following example to understand how we do use the departmental rates

A company uses two production departments and three service departments to manufacture heavy lifting gear for the offshore oil industry. After a detailed examination of costs and activities the company's management consultants produced the following estimates for next year.

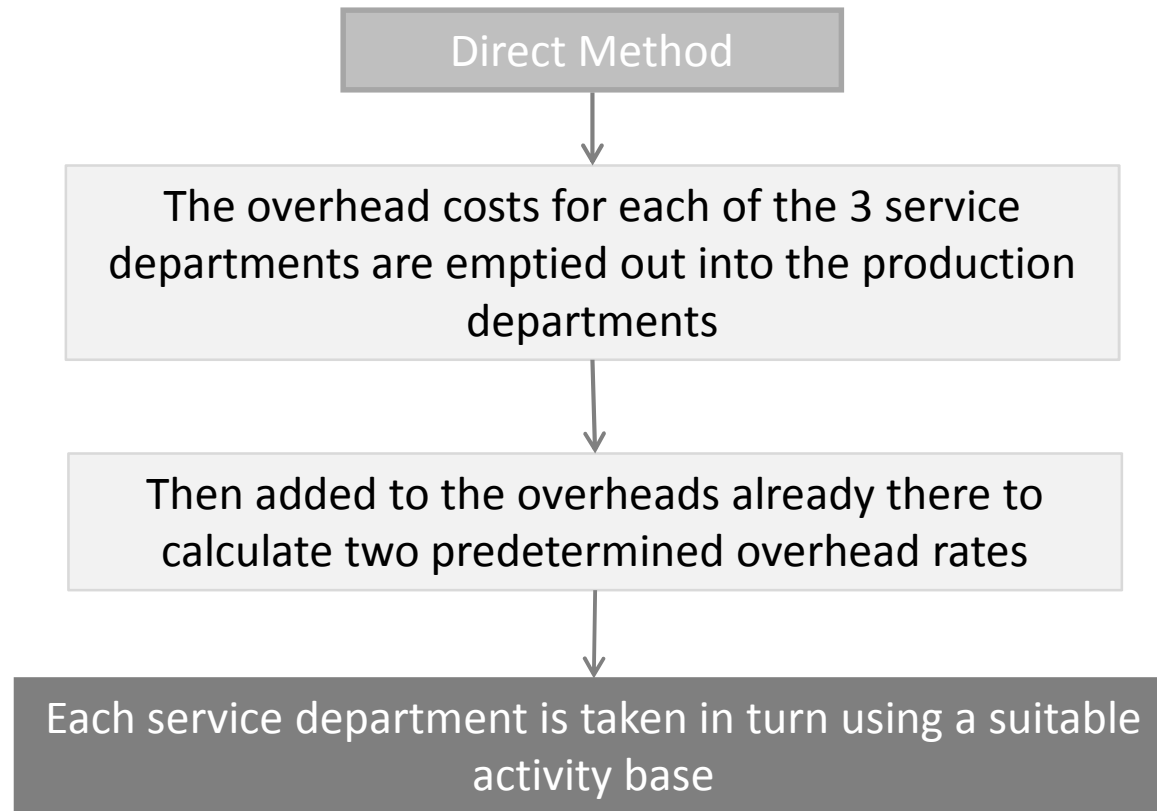
|                  | Overhead €     | Square metres occupied | No. of employees | Service hours |
|------------------|----------------|------------------------|------------------|---------------|
| Forging          | 252 500        | 4 500                  | 50               | 450           |
| Welding          | 113 750        | 3 000                  | 100              | 75            |
| Personnel        | 218 000        | 1 800                  | 30               | 225           |
| Computing        | 100 750        | 12 000                 | 20               | 4 500         |
| Site maintenance | 73 500         | 16 500                 | 10               | 1 500         |
|                  | <u>758 500</u> | <u>37 800</u>          | <u>210</u>       | <u>6 750</u>  |

In addition, it is forecast that there will be 45,000 machine hours within the Forging activity and 100,000 machine hours within the Welding activity.

**The first step in the allocation process is to transfer service department overheads to the production departments so that they can be 'attached' to the products**



## The direct method of overhead cost allocation is a straightforward approach to emptying out service department overhead costs



### Personnel

# of employees (Forging, 50; Welding, 100)

### Computing

Service hours (Forging, 450; Welding, 75)

### Site maintenance

Square meters occupied (Forging, 4500; Welding, 3000)

...So that we produce this comprehensive table

|                              |          |          |           |           |                  |          |
|------------------------------|----------|----------|-----------|-----------|------------------|----------|
|                              | Forging  | Welding  | Personnel | Computing | Site maintenance | Total    |
| Initial allocation           | €252 500 | €113 750 | €218 000  | €100 750  | €73 500          | €758 500 |
| Personnel<br>(Note 1)        | 72 667   | 145 333  | (218 000) |           |                  | 0        |
| Computing<br>(Note 2)        | 86 357   | 14 393   |           | (100 750) |                  | 0        |
| Site maintenance<br>(Note 3) | 44 100   | 29 400   |           |           | (73 500)         | 0        |
|                              | €455 624 | €302 876 | € 0       | € 0       | € 0              | €758 500 |

$= 50/150 \times €218,000$   
 $= 100/150 \times €218,000$   
 $= 450/525 \times €100,750$   
 $= 4500/7500 \times €73,500$   
 $= 75/525 \times €100,750$   
 $= 3000/7500 \times €73,500$

The consultants also studied the activities in Forging and Welding and recommended a machine-hours activity rate for Forging and a direct labor hours rate for Welding

$$\begin{aligned}\text{Predetermined overhead rates} &= \frac{\text{€455 624}}{45\,000} \text{ (machine hours)} \\ &= \text{€10.125 per machine hour}\end{aligned}$$

Every order which goes through Forging will be allocated €10.125 overhead for each machine-hour spent on the order in Forging

$$\begin{aligned}\text{Welding} & \frac{\text{€302 876}}{100\,000} \text{ (direct labour hours)} \\ & \text{€3.029 per direct labour hour}\end{aligned}$$

When the order moves to Welding, overhead of €3.029 will be allocated for every direct labor hour spent on the order

## Some companies prefer to recognize the interdepartmental services provided by each service department before the ultimate allocation to production department

### Example

In the company described before, Personnel renders a service not only to Forging and Welding but also to Computing and Site maintenance; Computing and Site maintenance similarly give a service to their sister service departments.

The ***step method***, recognizes this interdependency among departments



2 sequences

Select service departments in descending order of magnitude of overhead spend

We adopt this sequence

Select first the department that renders the highest percentage of its total services to other service departments and end with the one that renders the lowest percentage of service to other service departments

## The Step Method

Once a service department has been closed down (Personnel) no further service department's allocations are given to it. Therefore the number of service hours used to allocate Computing overhead is 2,025 (6,750 less those consumed by Computing and Personnel).

We ignore consumption of its own resources. Thus the total activity base for allocating personnel overhead is 180 persons, not 210 persons

|                              | Forging  | Welding  | Personnel | Computing | Site maintenance | Total    |
|------------------------------|----------|----------|-----------|-----------|------------------|----------|
| Initial allocation           | €252 500 | €113 750 | €218 000  | €100 750  | €73 500          | €758 500 |
| Personnel<br>(Note 1)        | 60 556   | 121 111  | (218 000) | 24 222    | 12 111           | 0        |
|                              |          |          |           | 124 972   |                  |          |
| Computing<br>(Note 2)        | 27 772   | 4 628    |           | (124 972) | 92 572           | 0        |
|                              |          |          |           |           | €178 183         |          |
| Site maintenance<br>(Note 3) | 106 910  | 71 273   |           |           | (178 183)        | 0        |
|                              | €447 738 | €310 762 | 0         | 0         | 0                | €758 500 |

## We reach a machine-hours activity rate for Forging and a direct labor hours rate for Welding

Forging  
Predetermined overhead rates –  $\frac{€447\,738}{45\,000}$  (machine hours)  
= €9.949 per machine hour

Every order which goes through Forging will be allocated €9.949 overhead for each machine-hour spent on the order in Forging

Welding  
 $\frac{€310\,762}{100\,000}$  (direct labour hours)  
€3.108 per direct labour hour

When the order moves to Welding, overhead of €3.108 will be allocated for every direct labor hour spent on the order

## We'll note that there is a small difference between the two methods

Consider two jobs going through both production departments:

|         | Forging           | Welding                 |
|---------|-------------------|-------------------------|
| Job 101 | 50 machine-hours  | 150 direct labour hours |
| Job 247 | 150 machine-hours | 50 direct labour hours  |

Using the direct method for allocating overhead the jobs would receive the following amounts:

|         | Forging                                  | Welding                               | Total     |
|---------|------------------------------------------|---------------------------------------|-----------|
| Job 101 | $50 \times \text{€}10.125$<br>€506.25    | $150 \times \text{€}3.029$<br>€454.35 | €960.60   |
| Job 247 | $150 \times \text{€}10.125$<br>€1 518.75 | $50 \times \text{€}3.029$<br>€151.45  | €1 670.20 |

Using the step method for allocating overhead the jobs would receive the following amounts:

|         | Forging                                 | Welding                               | Total     |
|---------|-----------------------------------------|---------------------------------------|-----------|
| Job 101 | $50 \times \text{€}9.949$<br>€497.45    | $150 \times \text{€}3.108$<br>€466.20 | €963.65   |
| Job 247 | $150 \times \text{€}9.949$<br>€1 492.35 | $50 \times \text{€}3.108$<br>€155.40  | €1 647.75 |

## **Allocating indirect costs is a problem which is universally experienced across all sectors of the economy**

### **Defense Research Establishment**

**A defense research establishment determining the cost of an electronic device developed for a battlefield communications system**

### **University**

**A university endeavoring to establish the cost of producing a medical graduate as opposed to an accounting graduate**

### **Hospital**

**A hospital costing a bed in the intensive care ward and one in the long-stay geriatric wing**

### **Bus Company**

**A bus company calculating the cost of a city centre commuter route versus long distance tourist journeys**

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**In some industries, work is concentrated on the production of one product but other products are produced which may or may not have significant market potential**

| Industry  | Main product     | Additional products                        |
|-----------|------------------|--------------------------------------------|
| Oil       | Aircraft fuel    | Various qualities of petrol; raw chemicals |
| Dairy     | Top quality beef | Poorer cuts of beef; milk; cheese; hide    |
| Chemicals | Plastic          | Components for detergents, fertilizers     |
| Quarrying | Granite          | Slate, road stone, sand                    |

**A joint product must appear during the processes involved in producing the main product**

### **Joint Product**

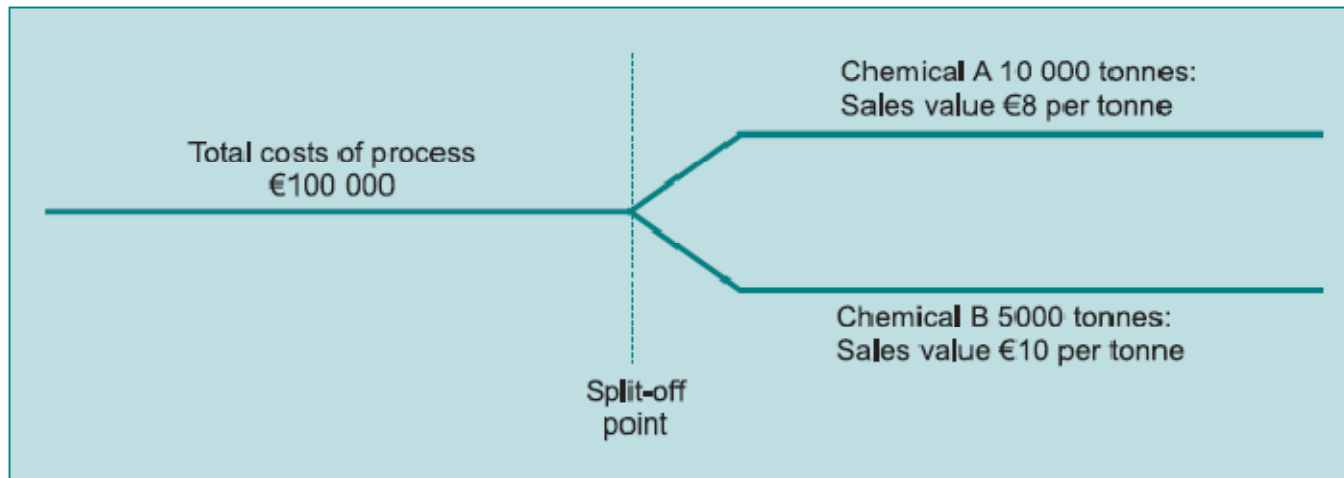
It must appear during the processes involved in producing the main product.  
If management has the option of not allowing the second product to emerge from the process the two products cannot be deemed to be joint.

**A question poses itself: how should the production costs be split over the two or more products which appear from the production process?**

**There are 4 joint-cost allocation methods that we'll explore here: 1) Equal shares; 2) Physical characteristics; 3) Sales value at split off; and 4) Ultimate Net Sales Value**

### **Example**

A chemical process produces Chemical A and Chemical B. Both can be sold at the point they emerge from production.



**1) Equal shares method, is driven by simplicity of operations and a desire to reflect the relative ease with which both products emerged from the production process**

|                        | Chemical A     | Chemical B | Total          |
|------------------------|----------------|------------|----------------|
| Sales value            | €80 000        | €50 000    | €130 000       |
| Joint production costs | 50 000         | 50 000     | 100 000        |
|                        | <u>€30 000</u> | <u>–</u>   | <u>€30 000</u> |

## 2) Physical characteristics method, assumes there is a simple linearity in the relationship between cost and cost characteristic (weight, volume, etc..)

We use weight here:

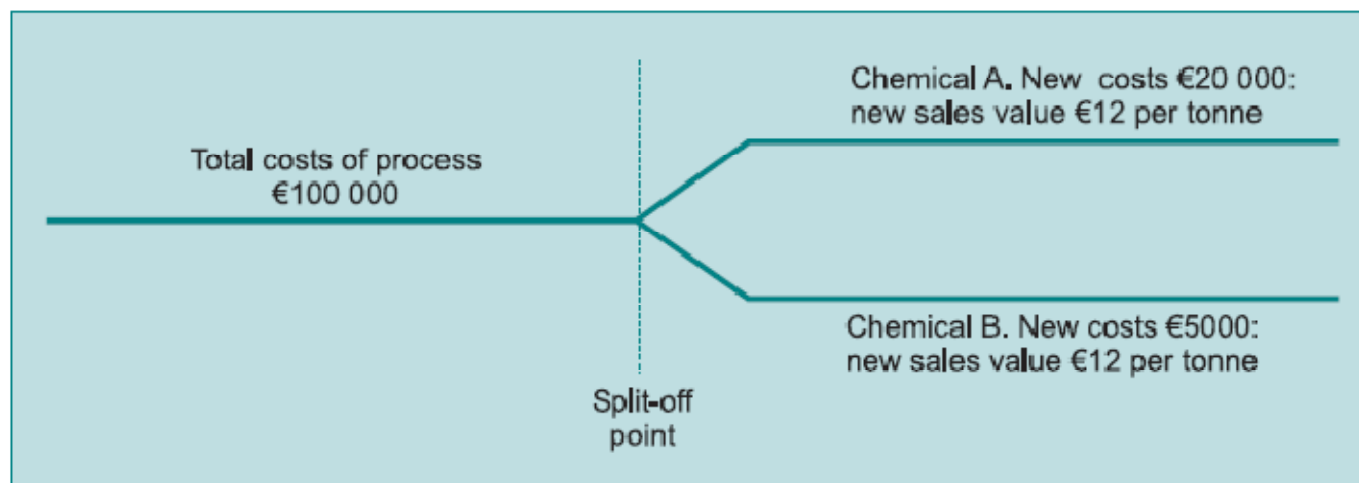
|                        | Chemical A     | Chemical B     | Total          |
|------------------------|----------------|----------------|----------------|
| Sales value            | €80 000        | €50 000        | €130 000       |
| Joint production costs | 66 667         | 33 333         | 100 000        |
|                        | <u>€13 333</u> | <u>€16 667</u> | <u>€30 000</u> |

**3) Sales value at split off method, considers the products' ability to bear the joint costs, which is being reflected in money generated by each product for the business**

|                        | Chemical A     | Chemical B     | Total          |
|------------------------|----------------|----------------|----------------|
| Sales value            | €80 000        | €50 000        | €130 000       |
| Joint production costs | 61 538         | 38 462         | 100 000        |
|                        | <u>€18 462</u> | <u>€11 538</u> | <u>€30 000</u> |

**4) Ultimate net sales value method, recognizes that the business may wish to process the 2 products further after the split-off point in an effort to maximize profit**

**Consider the new information given:**



**This method is considered an extension to the sales value at split off method**

The ultimate net sales value method deducts the new processing costs from the ultimate sales value and allocates the joint costs on the net amount

|                          | Chemical A | Chemical B | Total    |
|--------------------------|------------|------------|----------|
| Ultimate sales value     | €120 000   | €60 000    | €180 000 |
| New costs                | 20 000     | 5 000      | 25 000   |
| Ultimate net sales value | €100 000   | €55 000    | €155 000 |
| Joint production costs   | 64 516     | 35 484     | 100 000  |
|                          | €35 484    | €19 516    | €55 000  |

*Further processing costs of €20,000 can increase the sales price per tonne of chemical A by €4 (€12 less €8)*

*An increase of €2 per tonne is possible by a further expenditure of €5,000*

**Regardless of the allocation method selected, the results remain unchanged, so why should a business bother with such a complex and arbitrary procedure?**

**Several Reasons can Be Put Foreword:**

|   |                     |                                                                                                                                                                                             |
|---|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Product Management  | Decisions need to be made constantly on such matters as resources devoted to marketing, design & development of individual products.                                                        |
| 2 | Inventory Valuation | In order to construct the end-of-year external accounts for shareholders and other users, the business must attribute a value to the cost of goods sold and the cost of goods unsold        |
| 3 | Price-Setting       | Businesses are allowed to charge a fixed mark-up on proven cost of production. If the product in question is a joint product some attempts must be made to determine the cost of production |

## Test your Understanding of Inventory Valuation

Assume that the business could sell half the production of Chemical A and all of the production of Chemical B at the split-off point. Which joint cost-allocation method would produce the highest reported profit? (Use the original data.)

The answer is the physical characteristics method because Chemical A is allocated more cost under this method (namely €66,667) than under the others. Unsold production leads to costs being held back in inventory which are therefore not written off as cost of goods sold in the P/L account

**By-products are different from joint products only in terms of motive and commercial value**

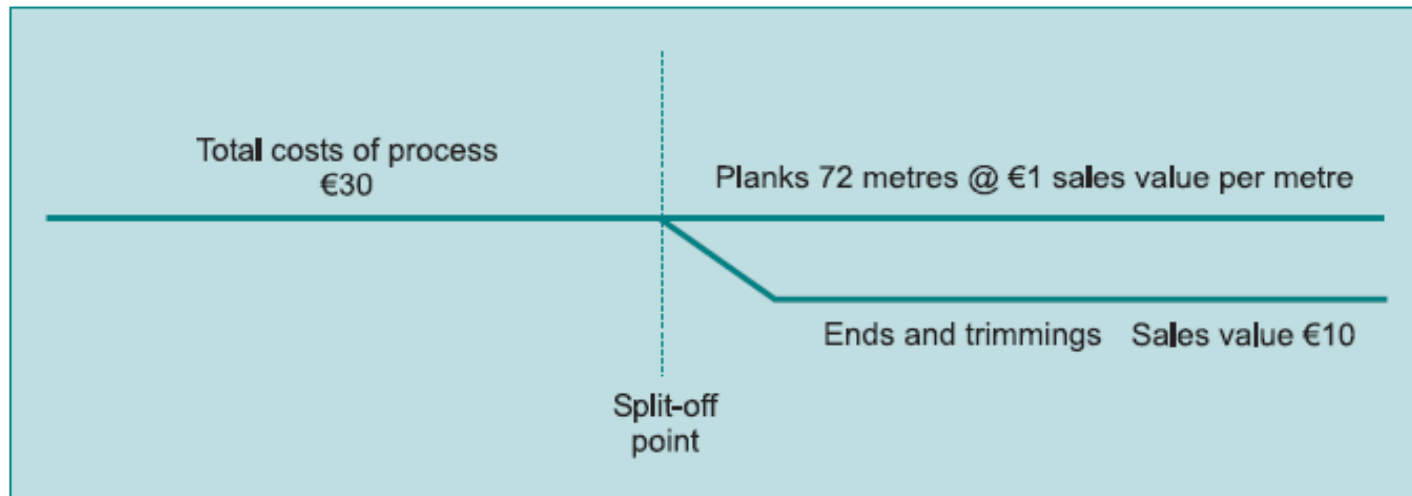
### **By-Product**

- A product which emerges from a production process that is designed to produce another product.
- Production and managerial effort is devoted to the main product but another, subsidiary, product emerges at the same time which has a low sales value relative to the main product.

## By-products are different from joint products only in terms of motive and commercial value

### Example

Consider a timber operation designed to produce planks for floorboards and fencing purposes. Awkward lengths of tree trunk, wood with knots and trimmings inevitably emerge from the sawing process; these can be viewed as by-products.



**No attempt would be made to allocate the costs of production separately between the main product and the by-product**

From the sawing operations



72 meters in length of floor boarding and fencing wood is produced with a retail value of €72 in total

+

The awkward lengths and trimming can fetch €10 per tree processed

|                               |     |           |
|-------------------------------|-----|-----------|
|                               |     | €         |
| Sales revenue                 |     | 72        |
| Cost of sales:                |     |           |
| Production costs              | €30 |           |
| Less: Revenue from by-product | 10  | 20        |
| Gross margin                  |     | <u>52</u> |

## The status of by-products can change over time; where for instance it could be transformed into a joint product

### Example

- The timber yard manager could receive an order from a local craft centre for timber trimmed ready for carving into tourist items. Modest trimming costs could enhance the sales value of the ends and trimmings.
- In such a situation the by-product may be elevated to the status of a joint product, depending on the relative sales values.
- Or the byproduct method of accounting may continue to be used when, from the revenue generated from the by-product, the additional costs of processing are deducted before reducing the total production costs by the net amount of revenue raised from the by-product.

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**"Job costing" is used where the units of output are identifiable, if they are not identifiable then we use "Process Costing"**

### Job Costing

The units of output are identifiable

The construction, aerospace, shipbuilding and consultancy industries

The outputs can be batched in larger units for costing purposes

### Process Costing

The units of output are **NOT** identifiable

The chemical industry or the brewing industry

The units of output are indistinguishable from each other

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**In job costing it's always hard to measure the units produced in an accounting period; that's why we use the notion of equivalent units of production**

### **Equivalent Units of Production**

An assessment of the degree of completion of a unit under each major component of cost

## Examples of equivalent units of production

### Example 1

- At the end of the financial year a business had completed the production of 12 million liters of paint. On the last day the production director assessed that 4 million liters were in production but that they were one-quarter finished. The equivalent units of production amount to:

$$12 \text{ million} + \frac{4 \text{ million}}{4} = 13 \text{ million equivalent units}$$

### Example 2

- Assume the same facts as above except that of the 12 million liters completed during the year, the production director reckoned that 2 million liters were in the opening work-in-process at the beginning of the year, each liter being half-finished. How many equivalent units were produced during the year?

$$\frac{2 \text{ million}}{2} + 10 \text{ million started and completed} + \frac{4 \text{ million}}{4} = 12 \text{ million}$$

*The logic behind it:*

If 12 million liters were completed during the year but 2 million of these were started in the previous year then only 10 million were started and completed in the year

Additionally, finishing 2M liters which were already 50% is equivalent to starting and completing 1M liters, and starting 4M liters to the extent of  $\frac{1}{4}$  completion = starting and completing another 1M units

## Examples of equivalent units of production (Continued)

### Example 3

- Reverting to the first scenario, imagine that all the chemicals required in the process of making paint were introduced at the start of the process and that direct labor and other conversion effort were applied evenly throughout the process.
- The equivalent unit calculation now has to be broken into two parts: material and labor/conversion.

|                                | Material         | Labour/conversion |
|--------------------------------|------------------|-------------------|
| Units started and completed    | 12 million       | 12 million        |
| Closing work-in-process        | <u>4 million</u> | <u>1 million</u>  |
| Equivalent units of production | 16 million       | 13 million        |

- These numbers indicate that the material input is complete even for the closing work-in-process but that labor and conversion is only one-quarter complete.
- When we attempt to calculate the cost of producing one liter of paint during this financial year, the material costs will be divided by 16 million and the labor and other conversion costs will be divided by 13 million

## Examples of equivalent units of production (Continued)

### Example 4

- Finally, reverting to the second scenario, imagine that opening work-in-process had all the material input added in the previous year and that only labor and conversion effort have to be made during the early days of this accounting period.

|                                | Material          | Labour/conversion |
|--------------------------------|-------------------|-------------------|
| Opening work-in-process        | –                 | 1 million         |
| Units started and completed    | 10 million        | 10 million        |
| Closing work-in-process        | 4 million         | 1 million         |
| Equivalent units of production | <u>14 million</u> | <u>12 million</u> |

- An equivalent unit is an assessment of the degree of completion of a unit under each major component of cost. Take the cost (as yet unknown) of labor and conversion effort during the accounting period; this cost has been applied to three blocks of production: (a) the completion of the 2 million liters of half finished opening work-in-process, (b) the 10 million liters started and completed during the year, and (c) the commencement of work on the 4 million liters lying in production at the end of the financial year, each liter of which was one-quarter finished in terms of labor and conversion.

## Self-Test on Equivalent Units

### Try the following example

A department in a multi-process factory received during June 10,000 units from the department which precedes it in the production process. At the beginning of the month its own work-in-process was assessed at 2,000 units (one-quarter finished in terms of material; three-quarters finished in terms of labor/conversion). By the end of the month 3000 units remained in production as closing work-in-process (one-half finished in terms of material; one-quarter in terms of labor/conversion). Calculate the equivalent units of production for the two major headings of production cost.

### Answer to Self-Test on Equivalent Units

|                                | Materials     | Labour/conversion |
|--------------------------------|---------------|-------------------|
| Opening work-in-process        | 1 500         | 500               |
| Units started and completed    | 7 000         | 7 000             |
| Closing work-in-process        | 1 500         | 750               |
| Equivalent units of production | <u>10 000</u> | <u>8 250</u>      |

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**The costs to be accounted for by a production activity in any accounting period comprise two parts**

**1**

***Beginning Work-in-process***

Costs attaching to opening work-in-process at the beginning of the period but which were incurred in the previous period

**2**

***Completing Work-in-process***

Costs incurred during this accounting period in order to finish the opening work-in-process

**The previous section dealt with the calculation of equivalent units; this section considers the application of cost to these units**

**To apply costs to equivalent units we use Scenario No. 4 from the previous section and repeat the background for ease of reference**

### **Example**

A business manufacturing paint started the year with 2 million liters partially finished opening work-in-process (complete as to materials; half-complete as to labor and conversion effort). During the year it started and completed 10 million liters and a further 4 million liters were in production at the year-end (complete as to materials and one-quarter complete in terms of labor and conversion effort).

The equivalent unit statement would be calculated as follows:

|                                | <b>Material</b><br>(millions of litres) | <b>Labour/conversion</b><br>(millions of litres) |
|--------------------------------|-----------------------------------------|--------------------------------------------------|
| Opening work-in-process        | –                                       | 1                                                |
| Units started and completed    | 10                                      | 10                                               |
| Closing work-in-process        | 4                                       | 1                                                |
| Equivalent units of production | <u>14</u>                               | <u>12</u>                                        |

The costs for the year can also be listed under the two headings of material and labor/conversion, conversion is assumed to be all direct & indirect overhead

|                          |                                             |                      |                       |             |
|--------------------------|---------------------------------------------|----------------------|-----------------------|-------------|
| Opening                  |                                             | Material             | Labour/<br>conversion | Total       |
|                          |                                             | €m                   | €m                    | €m          |
|                          | Value of opening work-in-process            | 0.8                  | 0.9                   | 1.7         |
|                          | Incurred during the year                    | 7.0                  | 12.0                  | 19.0        |
|                          | Total costs to account for                  | <u>7.8</u>           | <u>12.9</u>           | <u>20.7</u> |
| +                        |                                             |                      |                       |             |
| Current<br>Period        |                                             | Material             | Labour/conversion     |             |
|                          |                                             | (millions of litres) | (millions of litres)  |             |
|                          | Opening work-in-process                     | –                    | 1                     |             |
|                          | Units started and completed                 | 10                   | 10                    |             |
|                          | Closing work-in-process                     | 4                    | 1                     |             |
|                          | Equivalent units of production              | <u>14</u>            | <u>12</u>             |             |
| =                        |                                             |                      |                       |             |
| Direct Costs<br>Per Unit |                                             | Material             | Labour/<br>conversion |             |
|                          |                                             |                      |                       |             |
|                          | Costs for current period (from above)       | €7.0                 | €12.0                 |             |
|                          | Equivalent units of production (from above) | 14.0                 | 12.0                  |             |
|                          | Cost per equivalent unit                    | €0.50                | €1.0                  |             |
|                          | Total cost per equivalent unit              |                      | €1.50                 |             |

Then we allocate costs to production (i.e., allocating money values to the equivalent units statement using costs per equivalent unit)

Direct Costs  
Per Unit

|                                             |                  |                                |
|---------------------------------------------|------------------|--------------------------------|
| Costs for current period (from above)       | Material<br>€7.0 | Labour/<br>conversion<br>€12.0 |
| Equivalent units of production (from above) | 14.0             | 12.0                           |
| Cost per equivalent unit                    | €0.50            | €1.0                           |
| Total cost per equivalent unit              | €1.50            |                                |

X

Quantities

|                                | Material<br>(millions of litres) | Labour/conversion<br>(millions of litres) |
|--------------------------------|----------------------------------|-------------------------------------------|
| Opening work-in-process        | –                                | 1                                         |
| Units started and completed    | 10                               | 10                                        |
| Closing work-in-process        | 4                                | 1                                         |
| Equivalent units of production | <u>14</u>                        | <u>12</u>                                 |

=

Equiv. Cost  
Per Unit

|                             | Material<br>€m | Labour/<br>conversion<br>€m | Total<br>€m |
|-----------------------------|----------------|-----------------------------|-------------|
| Opening work-in-process     | –              | 1.0                         | 1.0         |
| Units started and completed | 5.0            | 10.0                        | 15.0        |
| Closing work-in-process     | 2.0            | 1.0                         | 3.0         |
|                             | <u>7.0</u>     | <u>12.0</u>                 | <u>19.0</u> |

## The valuation of units shipped out and closing inventory is then calculated

Equiv. Cost  
Per Unit



COGS & Inv.  
Valuation

|                             | Material   | Labour/<br>conversion | Total       |
|-----------------------------|------------|-----------------------|-------------|
|                             | €m         | €m                    | €m          |
| Opening work-in-process     | –          | 1.0                   | 1.0         |
| Units started and completed | 5.0        | 10.0                  | 15.0        |
| Closing work-in-process     | 2.0        | 1.0                   | 3.0         |
|                             | <u>7.0</u> | <u>12.0</u>           | <u>19.0</u> |

|                                                     |             |
|-----------------------------------------------------|-------------|
| Units shipped out                                   | €m          |
| Previously incurred cost of opening work-in-process | 1.7         |
| Costs to complete opening work-in-process           | 1.0         |
| Units started and completed                         | 15.0        |
|                                                     | <u>17.7</u> |
| Units left in closing work-in-process               | 3.0         |
| Total costs accounted for                           | <u>20.7</u> |

## Example of Weighted Average Process Costing; where similar steps will be followed

### Equivalent Units Statement

|                                | Material<br>(millions of litres) | Labour/ conversion<br>(millions of litres) |
|--------------------------------|----------------------------------|--------------------------------------------|
| Opening work-in-process        | 2                                | 2                                          |
| Units started and completed    | 10                               | 10                                         |
| Closing work-in-process        | 4                                | 1                                          |
| Equivalent units of production | <u>16</u>                        | <u>13</u>                                  |

**Here we ignore that some of our production effort was expended in the previous period and assume that it was all carried out during the current period**

**First we calculate the costs to be accounted for through adding value of opening work-in-progress and that incurred during the year**

**Costs to be accounted for**

|                                   | <b>Material</b> | <b>Labour/<br/>conversion</b> | <b>Total</b> |
|-----------------------------------|-----------------|-------------------------------|--------------|
|                                   | €m              | €m                            | €m           |
| Value of opening work-in-progress | 0.8             | 0.9                           | 1.7          |
| Incurred during the year          | 7.0             | 12.0                          | 19.0         |
|                                   | <u>7.8</u>      | <u>12.9</u>                   | <u>20.7</u>  |

**This time we assume the entire sum of €20.7 million was expended in the current year.**

**Then we divide the costs of production by the number of equivalent units of production to reach the total cost per equivalent unit**

**Cost per equivalent unit**

|                                | <b>Material<br/>(millions)</b> | <b>Labour/ conversion<br/>(millions)</b> |
|--------------------------------|--------------------------------|------------------------------------------|
| Costs of production            | €7.8                           | €12.9                                    |
| Equivalent units of production | 16                             | 13                                       |
| Cost per equivalent unit       | €0.4875                        | €0.9923                                  |
| Total cost per equivalent unit | €1.4798                        |                                          |

**Then we allocate costs to production (i.e., allocating money values to the equivalent units statement using costs per equivalent unit)**

|                             | Material      | Labour/<br>conversion | Total          |
|-----------------------------|---------------|-----------------------|----------------|
|                             | €m            | €m                    | €m             |
| Opening work-in-process     | 0.9750        | 1.9846                | 2.9596         |
| Units started and completed | 4.8750        | 9.9230                | 14.7980        |
| Closing work-in-process     | 1.9500        | 0.9924                | 2.9424         |
|                             | <u>7.8000</u> | <u>12.9000</u>        | <u>20.7000</u> |

## Finally we reach the valuation of units shipped out and closing inventory

### Value of units shipped out and closing inventory

|                                 |         |
|---------------------------------|---------|
| Units shipped out               | €m      |
| Opening work-in-process         | 2.9596  |
| Units started and completed     | 14.7980 |
|                                 | <hr/>   |
|                                 | 17.7576 |
| Units left in closing inventory | 2.9424  |
| Total costs accounted for       | <hr/>   |
|                                 | 20.7000 |

**The comparison between FIFO and the weighted average method shows that total process costing of both is the same**

**Comparison of FIFO and weighted average method of process costing**

|                  | Cost per<br>equivalent unit | Cost of units<br>shipped out | Cost of<br>closing work<br>in process | Total   |
|------------------|-----------------------------|------------------------------|---------------------------------------|---------|
|                  | €                           | €m                           | €m                                    | €m      |
| FIFO             | 1.5000                      | 17.7000                      | 3.0000                                | 20.7000 |
| Weighted average | 1.4798                      | 17.7576                      | 2.9424                                | 20.7000 |

The example used above displays some worrying signals which only the FIFO method picks up

A simple calculation will reveal that the costs per equivalent unit of the closing work-in-process last period were €0.40 for material and €0.90 for labor and conversion.

The FIFO method which calculates a period's costs precisely as incurred and without averaging in the previous period's displays respective unit costs as €0.50 and €1.00. These costs represent jumps of 25% and 11.11% respectively in the two sets of production costs

**If these rises represent anything more than a seasonal jump then management must be made aware of the situation, and only FIFO spells out the blunt message**

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**Activity-based costing focuses on the belief that activities, rather than products, cause costs to be incurred and products consume activities (costs)**

**Decoupling of costing from financial accounting**

ABC causes the cost of products to become the principal goal and not to be subservient to inventory valuation

**Different product, different cost drivers**

A business with complex production facilities employs multiple cost drivers => costs fall where they are consumed

**All overhead costs are embraced**

Such as R&D, quality control, production & marketing, and does not restrict itself to the production process' overhead

**Allowing for complex cost drivers**

ABC reflects the complexity of production & marketing permitting those products to be loaded up with those costs they incurred

**Strategic insight**

Providing accurate product costs; which may lead to strategic shifts in products and/or markets

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## It was found that in the sophisticated manufacturing environment ABC information is more helpful than traditional allocated accounting data to determine product costs

### Example

Take product diversity as an example of the changing sophisticated manufacturing environment : a company which wishes to increase the diversity of its products will inevitably increase the machine set-up costs and associated information technology costs

#### ABC Costing

Activity-based costing focuses on **cost drivers** – those activities or factors which generate cost



ABC identifies measurable activity like the number of suppliers, the number of machine set-ups, or the hours of associated computer runs

#### Traditional Costing

A traditional cost accounting system would treat these costs as part of manufacturing and/or administrative overhead and allocate them to the product using one of many allocation bases, such as direct labor hours, direct labor cost or machine-hours.

**These new costs have been caused neither by direct labor nor by machine-hours; rather their increase can be traced directly to the decision to introduce product diversity**

## Let's apply ABC costing to a real-life example...

### Example

Scottish River plc manufactures four products, Forth, Tay, Spey and Tweed using the same plant and equipment and similar production processes. The products differ dramatically in volume and production characteristics as follows.

| Product | Quantity<br>manufactured | Units of<br>material | Direct labour<br>hours per unit | Machine-hours<br>per unit |
|---------|--------------------------|----------------------|---------------------------------|---------------------------|
| Forth   | 10                       | 8                    | 1                               | 10                        |
| Tay     | 100                      | 6                    | 2                               | 6                         |
| Spey    | 1 000                    | 4                    | 3                               | 4                         |
| Tweed   | 10 000                   | 2                    | 4                               | 1                         |

## The 4 products use corporate resources such as direct material, direct labor, machine hours, quality control, computing centre and training in greatly varying proportions

The financial values and actual product usage for each product for one month are given below

|                   | Units of<br>material | Direct<br>labour<br>hours | Machine-<br>hours | Quality<br>control | Computing<br>centre | Training |
|-------------------|----------------------|---------------------------|-------------------|--------------------|---------------------|----------|
|                   |                      |                           |                   | %                  | %                   | %        |
| Forth             | 80                   | 10                        | 100               | 70                 | 40                  | 10       |
| Tay               | 600                  | 200                       | 600               | 10                 | 30                  | 10       |
| Spey              | 4 000                | 3 000                     | 4 000             | 10                 | 20                  | 10       |
| Tweed             | 20 000               | 40 000                    | 10 000            | 10                 | 10                  | 70       |
|                   | 24 680               | 43 210                    | 14 700            | 100%               | 100%                | 100%     |
| Value of Resource |                      |                           | €58 800           | €10 000            | €10 000             | €10 000  |

**Traditional costing systems might aggregate all overhead costs into one cost pool and allocate them to the four products using, perhaps, direct labor hours**

|                                            |                              |                          |
|--------------------------------------------|------------------------------|--------------------------|
| Total overhead to be allocated             |                              | €88 800                  |
| Allocation base: direct labour hours (DLH) |                              | 43 210                   |
| Overhead allocation per DLH                |                              | 2.055                    |
| <b>Product</b>                             | <b>DLH × Allocation rate</b> | <b>Overhead per unit</b> |
| Forth                                      | 1 × €2.055                   | €2.055                   |
| Tay                                        | 2 × €2.055                   | 4.110                    |
| Spey                                       | 3 × €2.055                   | 6.165                    |
| Tweed                                      | 4 × €2.055                   | 8.220                    |

*Tweed, is asked to bear four times the overhead of Forth because it consumes four times the amount of direct labor*



*If the individual items comprising overhead – machine-hours, quality control, computing centre and training – are not 'driven' by production volume these allocations will distort the ultimate product costs*

**An activity-based costing approach to this problem would be to identify the activities which contribute towards the delivery of the final products**

Time spent on the production machines

Inspection and other quality control procedures involved from the purchase of bought-in parts to final delivery to the customer

The use made by the overall production facilities of the computing centre

The training requirements of all personnel associated with the four products

**These activities are not necessarily related to volume of production; for example, product Forth is low volume but demanding in terms of quality control (70% of the total activity) and the computing centre (40% of the total activity)**

## We can now draw the cost patterns using activity-based costing

| Product | Machine-hours | Quality control | Computing centre | Training  | Total   |
|---------|---------------|-----------------|------------------|-----------|---------|
| Forth   | €400          | €7 000          | €4 000           | €1 000 =  | €12 400 |
| Tay     | 2 400         | 1 000           | 3 000            | 1 000 =   | 7 400   |
| Spey    | 16 000        | 1 000           | 2 000            | 1 000 =   | 20 000  |
| Tweed   | 40 000        | 1 000           | 1 000            | 7 000 =   | 49 000  |
|         | €58 800       | €10 000         | €10 000          | €10 000 = | €88 800 |

|       | Total overhead ÷<br>per product | Units of production | = | Overhead per unit |
|-------|---------------------------------|---------------------|---|-------------------|
| Forth | €12 400                         | 10                  |   | €1 240            |
| Tay   | 7 400                           | 100                 |   | 74                |
| Spey  | 20 000                          | 1 000               |   | 20                |
| Tweed | 49 000                          | 10 000              |   | 4.90              |

## A comparison between the two costing systems reveals a dramatic difference in the overhead portion of unit product costs

|       | Traditional<br>per DLH | Activity-based |
|-------|------------------------|----------------|
| Forth | €2.055                 | €1240          |
| Tay   | 4.110                  | 74             |
| Spey  | 6.165                  | 20             |
| Tweed | 8.220                  | 4.90           |

The conventional costing system consistently understates low-volume products (e.g. Forth) and overstates high-volume products (e.g. Tweed). The distortion in product costs (admittedly extreme in this illustration) caused by cross-subsidization could lead to strategic decisions which could harm the long-term interests of the enterprise. For instance, a decision based on traditional costing allocations to introduce further low-volume products in the mistaken belief that they are cheap to produce, or a decision to increase the sales price of high-volume products in a mistaken effort to recover higher overhead could inflict damage to the company from which recovery may be difficult.

## Case study

**The managing director of a local company that manufactures and sells high-quality paperweights speaks to you one day at the golf club:**

You ask how things are going! I haven't a clue. On the one hand we're selling one range of paperweights like hot cakes for a handsome price but our competitors accuse us of dumping below cost. On the other hand another range isn't selling well even though we've shaved our prices as far as we dare go. Even then the competition thinks we're overpricing and cutting our own throat. I should be grateful if you would come in to look at our cost system. Since I took over from my father we've relied on the cost accountant who is due to retire next year but I suspect he's hopelessly out of date.

**On your first visit you elicit the following information for the previous month from the cost accountant**

| Products                       | Islay  | Mull  | Skye  | Total    |
|--------------------------------|--------|-------|-------|----------|
| Monthly output (units)         | 10 000 | 8 000 | 3 000 |          |
| Direct material cost per unit  | €18    | €15   | €8    | €324 000 |
| Direct material usage per unit | 3      | 3     | 8     |          |
| Direct labour cost per unit    | €6     | €8    | €5    |          |
| Direct labour hours per unit   | 2      | 3     | 1     | 47 000   |

**The monthly overheads, listed below, are currently allocated to the three products on the basis of direct labor hours**

|                                | €               |
|--------------------------------|-----------------|
| Machine set-up                 | 20 000          |
| Machine energy and maintenance | 182 000         |
| Production scheduling          | 125 000         |
| Packing and shipping           | 61 000          |
| Design and development         | 80 000          |
|                                | <u>€468 000</u> |

## A plant tour enables you to watch the production process in detail and to speak to foremen and supervisors in each of the overhead departments

**To your question, ‘What is the one key feature of your activities that drives overhead costs?’, you get a variety of answers, none of which includes direct labor hours!**

Engineer-in-charge  
of production

‘Each switch from one line to another costs a packet. If we could cut down the number of switches my costs would fall.’ (Details on machine runs.)

Maintenance  
engineer

‘My costs are almost exclusively driven by machine hours as you would expect.’

Chief production  
scheduler

‘It’s sometimes thought that scheduling costs go up and down with output. But it is the number of orders we receive and schedule that drives our costs..’

Packing and  
shipping foreman

‘As you would expect it is the number of shipments made that drive our costs.’

Design Director

‘I’m not sure what our main cost driver is. At the end of the day I suppose it must be the number of patterns/designs accepted into production..’

The information gleaned from your plant tour can be summarized as follows

| Products                      | Islay | Mull | Skye | Total*  |
|-------------------------------|-------|------|------|---------|
| No. of production runs        | 3     | 6    | 18   | 27      |
| Machine-hours per unit        | 2     | 1    | 3    | 37 000* |
| No. of shipments              | 6     | 2    | 17   | 25      |
| No. of accepted designs       | 6     | 3    | 10   | 19      |
| No. of goods inwards orders** | 9     | 18   | 144  | 171     |

\* Comprising  $(2 \text{ hours} \times 10\,000) + (1 \times 8000) + (3 \times 3000)$

\*\* Direct material usage per unit  $\times$  No. of production runs

## Required

**On the basis of information received prepare a report for the managing director outlining the costs per unit of product range under the current scheme and suggest an alternative having regard to the actual cost drivers.**

## Analysis of case: We write a report with the following format, where we begin with calculating the overhead rate, then cost per product line

### Report to Managing Director of Paperweight plc

#### Need for activity-based costing

Having examined your costing system I am not surprised that you find a mismatch between your perception of your own costs and what the competition thinks is going on with your prices. Let me illustrate:

**Existing system.** Your current system, at least as elderly as your cost accountant and one that even your grandfather used, is based on spreading overhead to products on the basis of direct labour hours. On the previous month's costs (and I'm led to believe that last month's activities were typical of all other months in every regard) this procedure produces the following results:

$$\begin{aligned} 1 \text{ Overhead rate} &= \frac{\text{Total overhead}}{\text{Total direct labour hours}} \\ &= \frac{€468\,000}{47\,000} = €9.96 \text{ per DLH} \end{aligned}$$

| 2 Cost per product line | Islay        | Mull         | Skye         |
|-------------------------|--------------|--------------|--------------|
|                         | €            | €            | €            |
| Direct material         | 18.00        | 15.00        | 8.00         |
| Direct labour           | 6.00         | 8.00         | 5.00         |
| Share of overhead*      | 19.92        | 29.88        | 9.96         |
|                         | <u>43.92</u> | <u>52.88</u> | <u>22.96</u> |

\* Islay 2 × €9.96, Mull 3 × €9.96, Skye 1 × €9.96.

## Analysis of case (Continued)

I learn from your sales director that you seek to fix prices at a 50 per cent mark-up across all three ranges. This would imply that prices would be set at:

|                    | Islay  | Mull   | Skye   |
|--------------------|--------|--------|--------|
| Target sales price | €65.88 | €79.32 | €34.44 |
| Actual sales price | €65    | €55    | €35    |

And he also informed me that the two products you spoke to me about on the golf course were Mull (shaved prices to the lowest tolerable level but still the competition are wiping the floor with you in the market with much lower prices) and Skye (selling like hot cakes but the competitors accuse the company of dumping well below cost). I also notice that there is a marked quarterly decline in Skye sales and the sales director doubts that the success of the range can be taken for granted.

The use of direct labour hours is a blunt instrument. Overheads vary with a wide variety of factors, none of which is labour hours. This should not surprise you when you consider the low percentage of direct labour in the current cost profiles of the three products and the fact that so many other functions such as design and development and packing and shipping add value to the product with a very low labour base. On an activity-based costing approach each overhead is examined separately for the key 'driver' which causes costs to be incurred. The resulting cost profiles are very different from those above.

## Calculation of Cost of Overhead per Cost Driver

$$\text{Machine set-up costs} = \frac{€20\,000}{\text{No. of production runs}} = \frac{€20\,000}{27} = €740.74 \text{ per run}$$

$$\frac{\text{Machine energy and maintenance}}{\text{Machine-hours}} = \frac{€182\,000}{37\,000} = \frac{€182\,000}{37\,000} = €4.92 \text{ per machine-hour}$$

$$\frac{\text{Production scheduling}}{\text{No. of goods inwards orders}} = \frac{€125\,000}{171} = \frac{€125\,000}{171} = €731 \text{ per order}$$

$$\text{Packing and shipping} = \frac{€61\,000}{\text{No. of shipments}} = \frac{€61\,000}{25} = €2440 \text{ per shipment}$$

$$\text{Design and development} = \frac{€80\,000}{\text{No. of accepted designs}} = \frac{€80\,000}{19} = €4210 \text{ per design}$$

## Analysis of case (Continued)

| Overhead per unit calculation  |         | Islay                         | Mull                          | Skye                            |
|--------------------------------|---------|-------------------------------|-------------------------------|---------------------------------|
| Machine set-up costs           | €740.74 | $\times 3/10\,000$<br>= €0.22 | $\times 6/8\,000$<br>= €0.56  | $\times 18/3\,000$<br>= €4.44   |
| Machine energy and maintenance | €4.92   | $\times 2$<br>= €9.84         | $\times 1$<br>= €4.92         | $\times 3$<br>= €14.76          |
| Production scheduling          | €731    | $\times 9/10\,000$<br>= €0.66 | $\times 18/8\,000$<br>= €1.64 | $\times 144/3\,000$<br>= €35.09 |
| Packing and shipping           | €2 440  | $\times 6/10\,000$<br>= €1.46 | $\times 2/8\,000$<br>= €0.61  | $\times 17/3\,000$<br>= €13.83  |
| Design and development         | €4 210  | $\times 6/10\,000$<br>= €2.53 | $\times 3/8\,000$<br>= €1.58  | $\times 10/3\,000$<br>= €14.03  |

## Analysis of case (Continued)

| ABC cost profile       | Islay             | Mull              | Skye              |
|------------------------|-------------------|-------------------|-------------------|
|                        | €                 | €                 | €                 |
| Direct material        | 18.00             | 15.00             | 8.00              |
| Direct labour          | 6.00              | 8.00              | 5.00              |
| Share of overhead      |                   |                   |                   |
| Machine set-up         | 0.22              | 0.56              | 4.44              |
| Machine energy         | 9.84              | 4.92              | 14.76             |
| Production scheduling  | 0.66              | 1.64              | 35.09             |
| Packing and shipping   | 1.46              | 0.61              | 13.83             |
| Design and development | 2.53              | 1.58              | 14.03             |
|                        | <hr/> 38.71 <hr/> | <hr/> 32.31 <hr/> | <hr/> 95.15 <hr/> |

## Analysis of case (Continued)

ABC shows the company the basis of the comments being received from the competition

| Comparison of unit costs | Islay<br>€ | Mull<br>€ | Skye<br>€ |
|--------------------------|------------|-----------|-----------|
| Direct labour hours      | 43.92      | 52.88     | 22.96     |
| Activity-based costing   | 38.71      | 32.31     | 95.15     |

*The cost of manufacturing Mull falls well below current estimates and therefore management should contemplate a further price reduction to compete with other suppliers of similar quality*

*Paperweight's 'true' costs on Skye far exceed the current selling price of €35*

## ABC – A Concluding Note

### Cross-subsidization

Product diversity causes problems for traditional costing systems. The cost accounting system has to be designed to reflect the diversity otherwise the products with the low resource consumption will be asked to bear higher overhead costs than they deserve, and vice versa



*Reason behind it:*

The use of one cost pool and one cost driver, or over-aggregation of data

**If a business fails to examine the factors which drive the costs of discrete activities then the averages which emerge from the use of one rate will cause cross-subsidisation**

**A careful use of, say, two cost drivers could still produce a better picture of cost per unit than the use of one**

**We revert to the case and select machine-hours for the majority of the overhead but use the value of direct materials as a base for spreading production scheduling overhead**

$$\text{Production scheduling} = \frac{€125\,000}{€324\,000} = 38.58\% \text{ of direct material cost}$$

$$\text{Other overhead} = \frac{€468\,000 - €125\,000}{37\,000} = €9.27 \text{ per machine-hour}$$

| Two-cost centre cost profile | Islay        | Mull         | Skye         |
|------------------------------|--------------|--------------|--------------|
|                              | €            | €            | €            |
| Direct material              | 18.00        | 15.00        | 8.00         |
| Direct labour                | 6.00         | 8.00         | 5.00         |
| Share of overhead            |              |              |              |
| Production scheduling        | 6.94         | 5.79         | 3.09         |
| Other                        | 18.54        | 9.27         | 27.81        |
|                              | <u>49.48</u> | <u>38.06</u> | <u>43.90</u> |

**Although the costs of Mull and Skye are now moving towards their real ABC level they still are very far short of the numbers calculated under ABC, and Islay has moved in the wrong direction**

# Agenda

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10.1 Introduction

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10.2 Cost Gathering

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10.3 Where Do these Costs Come from?

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10.4 Plantwide versus Departmental Rates

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10.5 Joint Products and By-Products

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10.6 Process Costing

---

10.7 Process Costing and the Equivalent Unit

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10.8 Cost per Equivalent Unit

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10.9 Activity-Based Costing

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10.10 Traditional Costing versus ABC

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Review Questions

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Case Studies

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## Multiple Choice Questions

**10.1 In a cost-gathering system, which of the following costs is readily traceable to a cost item?**

- A. Sales travel expenses.
- B. Direct labor.
- C. Manufacturing overhead.
- D. Factory rent.

## Multiple Choice Questions (Continued)

**10.2 In pricing issues of materials from stores, it is possible to use different approaches, including:**

- i. the FIFO method;
- ii. the average method;
- iii. the replacement cost method;
- iv. the standard price method.

**Which of the following is correct?**

- A. (i) and (ii) only.
- B. (i), (ii) and (iii) only.
- C. (i), (ii) and (iv) only.
- D. (iii) and (iv) only.

## Multiple Choice Questions (Continued)

**10.3 The purpose of 'just-in-time' materials control systems is to ensure that cash tied up in inventories is maximized. True or false?**

False

## Multiple Choice Questions (Continued)

**10.4 Which of the following source documents is typically used by cost accountants to attach labor costs to specific jobs?**

- A. Weekly wages records.
- B. Monthly salary records.
- C. Employees clockcards.
- D. Daily time sheets.

## Multiple Choice Questions (Continued)

**10.5 In compiling a wage rate for use in cost gathering, cost accountants aggregate:**

- i. gross wages or salaries;
- ii. employees' social security costs;
- iii. employers' pension contributions;
- iv. employers' social security costs.

**Which of the following is correct?**

- A. (i) only.
- B. (i) and (ii) only.
- C. (i), (ii) and (iii) only.
- D. (i), (iii) and (iv) only.

## Multiple Choice Questions (Continued)

**10.6 Manufacturing overhead includes cost items which have a direct and close relationship with the output from the manufacturing activity. True or false?**

False

## Multiple Choice Questions (Continued)

**10.7 Which of the following is an example of a non-manufacturing overhead?**

- A. Factory heating and lighting.
- B. Depreciation of plant.
- C. Research and development expenditure.
- D. Maintenance of machinery.

## Multiple Choice Questions (Continued)

**10.8 Which of the following defines a predetermined overhead allocation rate?**

- A. Budgeted overhead divided by budgeted units of production.
- B. Actual overhead divided by budgeted units of production.
- C. Budgeted overhead divided by actual units of production.
- D. Actual overhead divided by actual units of production.

## Multiple Choice Questions (Continued)

**10.9 In allocating overheads to cost units, cost accountants seek to identify activity bases, which are causal forces in the incurrence of manufacturing overhead. Typically, these include:**

- i. direct labor costs;
- ii. direct machine-hours;
- iii. direct machine costs;
- iv. direct labor hours.

**Which of the following is correct?**

- A. (i) and (ii) only.
- B. (i), (ii) and (iii) only.
- C. (i), (ii) and (iv) only.
- D. (iii) and (iv) only.

## Multiple Choice Questions (Continued)

**10.10 Which of the following describes an over-application of overheads?**

- A. Actual overheads exceed budgeted overheads.
- B. Applied overheads exceed budgeted overheads.
- C. Actual overheads exceed applied overheads.
- D. Applied overheads exceed actual overheads.

## Multiple Choice Questions (Continued)

**10.11 Which of the following is the normal accounting treatment for an underapplication of overheads?**

- A. A reduction in work-in-progress.
- B. An addition to cost of goods sold.
- C. A reduction in cost of goods sold.
- D. An increase in work-in-progress.

## Multiple Choice Questions (Continued)

**10.12 Which of the following is the primary purpose of overhead allocation rates?**

- A. Reallocation of service centre costs to manufacturing departments.
- B. Measurement of interdepartmental performance.
- C. Application of manufacturing overheads into cost units.
- D. Comparison of actual overheads with budgeted overheads.

## Multiple Choice Questions (Continued)

**10.13 If manufacturing overhead is over-applied by €42,000 and applied overhead is €358,000, what is the amount of actual overhead?**

- A. €42,000.
- B. €316,000.
- C. €358,000.
- D. €400,000.

|                        |                 |
|------------------------|-----------------|
| Applied overhead       | €358 000        |
| Less: Over-application | 42 000          |
| Actual overhead        | <u>€316 000</u> |

## Multiple Choice Questions (Continued)

**10.14 Hunter and Howndes Limited has been operating with an overhead rate of €7.50 per direct labor hour. During the year, actual overheads totaled €315,000 and 41,000 direct labor hours were worked. Which of the following statements is correct?**

- A. The under-application of overhead was €6,400.
- B. The over-application of overhead was €6,400.
- C. The over-application of overhead was €7,500.
- D. The under-application of overhead was €7,500.

|                                   |              |
|-----------------------------------|--------------|
|                                   | €            |
| Applied overhead (41 000 × €7.50) | 307 500      |
| Actual overhead                   | 315 000      |
| Under-application                 | <u>7 500</u> |

## Multiple Choice Questions (Continued)

**10.15 In preferring departmental overhead rates to plantwide overhead rates, management is able to generate more accurate product costs by selecting different activity bases for different departments. True or false?**

True

## Case Study 10.1

**Leith Engineering Co. is introducing a system of allocating overheads to its output through the use of predetermined overhead rates. The management accountant has prepared the following estimates of budgeted expenditure for next year.**

| Department  | Allocated indirect costs<br>€ | Machine-hours |
|-------------|-------------------------------|---------------|
| Personnel   | 100 000                       | –             |
| Maintenance | 75 000                        | –             |
| Catering    | 60 000                        | –             |
| Fabrication | 300 000                       | 50 000        |
| Welding     | 200 000                       | 40 000        |

In addition, there are some indirect costs not yet allocated:

|                  | €      |
|------------------|--------|
| Rent and rates   | 24 000 |
| Power (machines) | 18 000 |
| Depreciation     | 15 000 |
| Heat and light   | 12 000 |

## Case Study 10.1 (Continued)

Further information is available:

| Department  | Number of employees | Plant value (€) | Area (square metres) |
|-------------|---------------------|-----------------|----------------------|
| Personnel   | 5                   | 4 000           | 5 000                |
| Maintenance | 10                  | 9 000           | 5 000                |
| Catering    | 10                  | 8 000           | 15 000               |
| Fabrication | 50                  | 75 000          | 20 000               |
| Welding     | 30                  | 54 000          | 15 000               |

The company has decided to opt for the step method of overhead allocation.

### Required

1. After treating unallocated costs, apply the step method of reallocation to compute the overhead rates for the manufacturing departments.
2. If a plantwide rate is used, what are its implications for product costing?

## Case Study 10.1 (Solution)

|                                           | Manufacturing departments |         | Service departments |             |                 | Total   |
|-------------------------------------------|---------------------------|---------|---------------------|-------------|-----------------|---------|
|                                           | Fabrication               | Welding | Personnel           | Maintenance | Catering        |         |
|                                           | €                         | €       | €                   | €           | €               | €       |
| Allocated costs                           | 300 000                   | 200 000 | 100 000             | 75 000      | 60 000          | 735 000 |
| Rent and rates                            |                           |         |                     |             |                 |         |
| – See Note (1)                            | 8 000                     | 6 000   | 2 000               | 2 000       | 6 000           | 24 000  |
| Power                                     |                           |         |                     |             |                 |         |
| – See Note (2)                            | 10 000                    | 8 000   | –                   | –           | –               | 18 000  |
| Depreciation                              |                           |         |                     |             |                 |         |
| – See Note (3)                            | 7 500                     | 5 400   | 400                 | 900         | 800             | 15 000  |
| Heat and light                            |                           |         |                     |             |                 |         |
| – See Note (4)                            | 4 000                     | 3 000   | 1 000               | 1 000       | 3 000           | 12 000  |
|                                           | 329 500                   | 222 400 | 103 400             | 78 900      | 69 800          | 804 000 |
| Reallocation                              |                           |         |                     |             |                 |         |
| – See Note (5)                            | 51 700                    | 31 020  | (103 400)           | 10 340      | 10 340          | –       |
|                                           | 381 200                   | 253 420 | –                   | 89 240      | 80 140          | 804 000 |
| Reallocation                              |                           |         |                     |             |                 |         |
| – See Note (6)                            | 49 578                    | 39 662  |                     | (89 240)    | –               | –       |
|                                           | 430 778                   | 293 082 |                     | –           | 80 140          | 804 000 |
| Reallocation                              |                           |         |                     |             |                 |         |
| – See Note (7)                            | 50 088                    | 30 052  |                     |             | (80 140)        | –       |
|                                           | 480 866                   | 323 134 |                     |             | –               | 804 000 |
| Machine-hours                             | 50 000                    | 40 000  |                     |             |                 | 90 000  |
| Overhead allocation rate per machine-hour | €9.617                    | €8.078  |                     |             | Plant-wide rate | €8.93   |

## Case Study 10.1 (Solution)

### Notes

1. Allocated on the basis of area occupied by each department out of total area of 60 000 square metres:

$$\text{e.g. Personnel } \frac{5\,000}{60\,000} \times \text{€}24\,000 = \text{€}2\,000$$

2. Allocated on the basis of machine-hours in the manufacturing departments out of 90 000 total machine-hours:

$$\text{e.g. Fabrication } \frac{50\,000}{90\,000} \times \text{€}18\,000 = \text{€}10\,000$$

3. Allocated on the basis of plant value in each department out of total plant value of €150 000:

$$\text{e.g. Welding } \frac{\text{€}54\,000}{\text{€}150\,000} \times \text{€}15\,000 = \text{€}5\,400$$

4. Allocated on the basis of area occupied by each department out of total area of 60 000 square metres:

$$\text{e.g. Maintenance } \frac{5\,000}{60\,000} \times \text{€}12\,000 = \text{€}1\,000$$

## Case Study 10.1 (Solution)

5. Personnel department is selected as the first service department for reallocation since it has the highest level of overhead expenditure. Overheads are reallocated on the basis of numbers employed by each department out of total employees of 100 (*excluding* Personnel):

$$\text{e.g. Maintenance } \frac{10}{100} \times \text{€}103\,400 = \text{€}10\,340$$

6. No information is available about the activities of the Maintenance Department. Therefore, it is necessary to adopt the machine-hours information and use it as a basis for allocation, as in Note (2) above.
7. As in Note (5) above, it appears that numbers of employees is the most suitable base for allocation of catering costs (excluding Catering and other service departments):

$$\text{e.g. Fabrication } \frac{50}{80} \times \text{€}80\,140 = \text{€}50\,088$$

2. A plantwide rate (of €8.93) would result in less accurate product costings. For example, any product requiring a high Fabrication content and low Welding content would be under-costed to a significant extent, €9.62 for Fabrication against a plantwide rate of €8.93.

## Case Study 10.2

Jagger Products Limited manufactures toy dolls in a molding process. The management accountant has been taken ill and you have been requested to assist in identifying the usual monthly adjustments to measure output and cost of output.

Opening work-in-process amounted to 8,000 dolls, with costs attached of materials €3,984 and labor/conversion of €3,840. Materials are added at the beginning of the molding process. Opening work-in-process is 50 per cent complete in terms of labor/conversion costs.

During the month, a further 40 000 dolls have been input to the process, incurring material costs of €24,000 and labor/conversion costs of €39,936. Completed dolls totaled 42,000 in the month, leaving 6000 in work-in-process, 60% complete in terms of labor/conversion costs.

The company operates a FIFO approach to accounting for stock movements.

### Required

Prepare the necessary computations to support the costs attached to:

- (a) closing work-in-process;
- (b) units started and completed;
- (c) completion of opening work-in-process.

## Case Study 10.2 (Continued)

### 1 Equivalent unit statement

|                             | Materials     | Labour/ conversion                 |                |
|-----------------------------|---------------|------------------------------------|----------------|
| Opening work-in-process     | –             | 4 000 (8000 units 50% to complete) |                |
| Units started and completed | 34 000        | 34 000                             |                |
| Closing work-in-process     | 6 000         | 3 600 (6000 units 60% completed)   |                |
| Equivalent units            | <u>40 000</u> | <u>41 600</u>                      |                |
| Costs to be accounted for   |               |                                    | Total          |
| Incurred in month           | €24 000       | €39 936                            | <u>€63 936</u> |
| Costs per equivalent unit   | <u>€0.60</u>  | <u>€0.96</u>                       |                |
|                             | <u>€1.56</u>  |                                    |                |

#### (a) Closing work-in-process

|                   |                |   |               |
|-------------------|----------------|---|---------------|
| Materials         | (6000 × €0.60) | = | €3 600        |
| Labour/conversion | (3600 × €0.96) | = | €3 456        |
|                   |                |   | <u>€7 056</u> |

#### (b) Units started and completed

$$(34\,000 \times €1.56) = \underline{€53\,040}$$

#### (c) Completion of opening work-in-process

|                   |                    |   |               |
|-------------------|--------------------|---|---------------|
| Materials         | (already complete) | = | –             |
| Labour/conversion | (4000 × €0.96)     | = | <u>€3 840</u> |

In completing opening work-in-process, costs incurred last month and brought forward (Materials €3,984 and Labor/conversion €3,840) will also be transferred to finished goods.

## Case Study 10.3

Precision Joinery Co. manufactures high-quality wooden products for the house building sector, specializing in window-frames, doors and kitchen cabinets. For many years, the company's costing system has relied on the assumption that direct labor hours were the critical factor in the incurrence of overheads. Accordingly, overheads were allocated to products on the basis of the direct labor hours taken to manufacture each product.

The company's relationship with its customers has changed in recent years. House building companies have become more rigorous in their demands in terms of both product design and service support after delivery. The Managing Director of Precision Joinery Co. has become increasingly disillusioned with the current product costing system, which he believes to be producing costs which do not reflect the change in the market environment initiated by the company's customers.

## Case Study 10.3 (Continued)

For May, the following information has been extracted from the current production costing system:

|                              | Window-frames | Doors | Cabinets |
|------------------------------|---------------|-------|----------|
| Sales (units)                | 2000          | 3000  | 1000     |
| Variable costs (per unit)    |               |       |          |
| labour                       | €20           | €35   | €25      |
| materials                    | €30           | €40   | €50      |
| Labour hours (per unit)      | 4             | 7     | 5        |
| Overheads for May = €155 000 |               |       |          |

Under duress, the Management Accountant has carried out a further examination of available costing information, some of which the Managing Director considers useful in a possible redefinition of the costing system.

**Overheads comprise:**

|                          | €       |
|--------------------------|---------|
| Materials handling costs | 20 000  |
| Batch set-up costs       | 12 000  |
| Product design           | 30 000  |
| Machining costs          | 45 000  |
| Service support          | 20 000  |
| Purchasing Department    | 28 000  |
|                          | <hr/>   |
|                          | 155 000 |

## Case Study 10.3 (Continued)

The Management Accountant's study has revealed the following product information:

|                          | Window-frames<br>(No.) | Doors (No.) | Cabinets<br>(No.) | Total |
|--------------------------|------------------------|-------------|-------------------|-------|
| Customer orders          | 15                     | 5           | 10                | 30    |
| Production batches       | 6                      | 12          | 6                 | 24    |
| Purchase orders          | 4                      | 3           | 14                | 21    |
| Receipts of material     | 5                      | 15          | 10                | 30    |
| Service visits           | 8                      | 4           | 3                 | 15    |
| Machine-hours (per unit) | 4                      | 6           | 8                 | 18    |

All units manufactured in May were sold.

## Case Study 10.3 (Continued)

### Required

The Management Accountant is unfamiliar with recent developments in product costing. He is vaguely aware of ABC costing and calls you in to progress its implementation in Precision Joinery Co.

Using the information available:

1. Select suitable cost drivers for overheads.
2. Compute the ABC product cost and compare it with the output of the traditional product costing system
3. If current selling prices are based on a 25 per cent mark-up of product cost, what will be the effect of the introduction of ABC approach on the assessment of product profitability?

## Case Study 10.3 (Solution)

1

1. From the information available, the following set of cost drivers with appropriate overheads can be identified:

| Overhead                 | Cost driver          |
|--------------------------|----------------------|
| Materials handling costs | Receipts of material |
| Batch set-up costs       | Production batches   |
| Product design           | Customer orders      |
| Machining costs          | Machine-hours        |
| Service support costs    | Service visits       |
| Purchasing Department    | Purchase orders      |

## Case Study 10.3 (Solution)

2. The company's traditional product costing system relies on an interaction between direct labour hours (DLH) and overheads. The higher the level of direct labour hours for a product, the larger its overhead cost content. The company's overhead allocation rate is:

$$\frac{\text{Overheads}}{\text{Direct labour hours}} = \frac{€155\,000}{34\,000} = €4.56 \text{ per direct labour hour}$$

|                           |   |               |            |               |
|---------------------------|---|---------------|------------|---------------|
| where direct labour hours | = | Window-frames | (4 × 2000) | 8 000         |
|                           |   | Doors         | (7 × 3000) | 21 000        |
|                           |   | Cabinets      | (5 × 1000) | 5 000         |
|                           |   |               |            | <u>34 000</u> |

### 'Traditional' product costs

|               | Window-frames | Doors         | Cabinets     |
|---------------|---------------|---------------|--------------|
|               | €             | €             | €            |
| Labour        | 20.00         | 35.00         | 25.00        |
| Materials     | 30.00         | 40.00         | 50.00        |
| Overheads     |               |               |              |
| 4 DLH × €4.56 | 18.24         | –             | –            |
| 7 DLH × €4.56 | –             | 31.92         | –            |
| 5 DLH × €4.56 | –             | –             | 22.80        |
|               | <u>68.24</u>  | <u>106.92</u> | <u>97.80</u> |

## Case Study 10.3 (Solution)

### ABC approach to product costs

Overheads are allocated on the basis of their relative consumption of the cost drivers, i.e. factors identified as causing their incurrence, as noted in 1 above, e.g. the number of customer orders per product influences the incurrence of product design costs of €30 000. With 30 customer orders in total and five received in respect of doors, the Doors overhead cost element is weighted with:

$$\frac{5}{30} \times €30\,000 = €5\,000$$

## Case Study 10.3 (Solution)

|                    | Window-frames | Doors         | Cabinets      | Total          | Cost driver          |
|--------------------|---------------|---------------|---------------|----------------|----------------------|
|                    | €             | €             | €             | €              | €                    |
| Materials handling | 3 333         | 10 000        | 6 667         | 20 000         | Receipts of material |
| Batch set-up       | 3 000         | 6 000         | 3 000         | 12 000         | Production batches   |
| Production design  | 15 000        | 5 000         | 10 000        | 30 000         | Customer orders      |
| Machining          | 10 588        | 23 824        | 10 588        | 45 000         | Machine-hours*       |
| Service support    | 10 667        | 5 333         | 4 000         | 20 000         | Service visits       |
| Purchasing Dept    | 5 333         | 4 000         | 18 667        | 28 000         | Purchase orders      |
|                    | <u>47 921</u> | <u>54 157</u> | <u>52 922</u> | <u>155 000</u> |                      |

|                       |               |            |               |
|-----------------------|---------------|------------|---------------|
| *Total machine-hours: | Window-frames | (4 × 2000) | 8 000         |
|                       | Doors         | (6 × 3000) | 18 000        |
|                       | Cabinets      | (8 × 1000) | 8 000         |
|                       |               |            | <u>34 000</u> |

Since window-frames require 8000 machine-hours out of 34 000 total machine-hours, the product line is allocated overhead in line with its machine-hour usage, as follows:

$$\frac{8\,000}{34\,000} \times \text{€}45\,000 = \text{€}10\,588$$

## Case Study 10.3 (Solution)

|                          | Window-frames | Doors  | Cabinets |
|--------------------------|---------------|--------|----------|
|                          | €             | €      | €        |
| Labour                   | 20.00         | 35.00  | 25.00    |
| Materials                | 30.00         | 40.00  | 50.00    |
| Overheads                |               |        |          |
| €47 921/2000             | 23.96         | –      | –        |
| €54 157/3000             | –             | 18.05  | –        |
| €52 922/1000             | –             | –      | 52.92    |
|                          | <hr/>         | <hr/>  | <hr/>    |
|                          | 73.96         | 93.05  | 127.92   |
| Traditional product cost | <hr/>         | <hr/>  | <hr/>    |
|                          | 68.24         | 106.92 | 97.80    |

The major differences between the traditional approach and ABC are emphasised in the higher-volume and lower-volume products, with the lower-volume product, cabinets, being asked to carry a much higher burden of overheads, €52.92 per unit instead of €22.80 per unit.

## Case Study 10.3 (Solution)

3.

|                             | Window-frames | Doors        | Cabinets      |
|-----------------------------|---------------|--------------|---------------|
|                             | €             | €            | €             |
| Current selling price       | 85.30         | 133.65       | 122.25        |
| (Traditional cost plus 25%) |               |              |               |
| ABC product cost            | <u>73.96</u>  | <u>93.05</u> | <u>127.92</u> |

The obvious conclusion is that, although the company is achieving a significantly higher margin than it realises on doors, it is selling cabinets at an overall loss of €5.67 per unit. Consideration should be given to either increasing the price of cabinets or the elimination of the product line and its replacement with more of the existing products or even a new product line.